



Faculty of Science and Technology

Programme Structure & Syllabus

For

Third Year

Bachelor of Science (Honors / Research)

Computer Science

(BSC CS)

P-2023

With Effect from Academic Year 2023– 2024

	Authority	Date
Proposed by	Board of Studies in Computer Science	
Approved by	Academic Council, Vishwakarma University, Pune	

Issued by

Chairman – BoS

Dean of Faculty

Director, IQAC

Title: Programme Structure and Syllabus

Form No: IQAC-101

Vision of the University	
Emerge as a Premier University Recognized Internationally for Excellence in Education, Research and Innovation	
Mission of the University	
VU-M1	To impart contemporary transformative education through research and innovation
VU-M2	To develop competent leaders-professionals for life and livelihood
VU-M3	To co-create human and socio-economic capital par excellence
VU-M4	To inculcate life skills and holistic culture appreciating morals and ethics
Values of the University	
Excellence	Transparency
Innovation	Sustainability
Diversity	Responsibility
Adaptability	Compassion

Vision of the Department of Computer Science	
Emerge as renowned research and innovation-oriented department in the field of Information Technology	
Mission of the Department of Computer Science	
M1	To impart industry centric education through research and innovation
M2	To develop skilled professionals to meet global socio-economic challenges
M3	To introduce best practices in software development adhering to moral and ethics.
M4	To foster collaborative approach and leadership skills.
Values of the Department of Computer Science	
Integrity	Brightness
Innovation	Diversity
Ethics	Competitive
Leadership	Lucidity

Programme Educational Objectives (PEOs)

PEO No.	Statement
PEO1	Graduates will have a thorough foundational knowledge of the principles and practices of Computer Science, and be proficient in applying them. (Based on Knowledge)
PEO2	To be able to adapt new tools/techniques to effectively contribute as an individual and lead the software development team to provide economically and technically viable solutions to a problem. (Based on Skills)
PEO3	Commit to ethical and moral practices for providing solutions to societal problems through life-long learning (Based on Attitude)

Mapping of Mission Statement of Department to PEOs

Mission Statement	PEO1	PEO2	PEO3
M1	3	2	3
M2	1	3	2
M3	1	2	3
M4	1	3	1

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Outcomes (POs)

PO No.	Statement
PO1	Possess profound knowledge of theory and principles of basic mathematics, statistical techniques and algorithms in the field of computer science.
PO2	Ability to think critically and select economically and technically viable solutions to a given problem.
PO3	Ability to design, develop and deploy solutions for identified problems from diverse domains.
PO4	Provide proof of concept and present novel solutions to challenges encountered by industry.
PO5	Adopt new programming languages, tools, and processes followed in industry
PO6	To work independently or as an effective team member of a software development project.
PO7	Effectively communicate with diverse clients and stakeholders for software project management.
PO8	Interpersonal skills to interact with people from diverse cultural backgrounds
PO9	Adhere to the professional, ethical and legal practices followed in the IT industry.
PO10	Possess software project management skills like: cost estimation, effort estimation, scheduling, and team building.

Mapping of PEOs to Pos

PEO Number	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
PEO1	3	2	3	2	1	-	-	-	-	-
PEO2	1	3	-	2	3	2	2	1	1	2
PEO3	-	-	2	-	-	1	1	2	3	1

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Specific Outcomes (PSOs)

PSO No.	Statement
PSO1	Choose appropriate data structure, algorithm, software development paradigm to address real life problems.
PSO2	Apply modern tools and techniques used in Data Science and Game Development.
PSO3	Provide the solutions to computing challenges from multidisciplinary fields.
PSO4	Possess a research and development attitude focusing on knowledge creation and innovation.

Mapping of PEOs to PSOs

PEO Number	PSO1	PSO2	PSO3	PSO4
PEO1	-	-	-	4
PEO2	2	2	1	-
PEO3	3	-	3	-
PEO4	-	-	-	4

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Structure

Faculty	Science and Technology	Pattern	2023
Department	Computer Science	Date (w.e.f.)	
Programme	B.Sc. – Computer Science TY	Semester	V

CCS Semester –V										
Cour se Type	Course Code	Course Name	Teaching Scheme (Hours/Week)			Credit	Examination Scheme and Marks			
			L	T	P	C	CIE	ESE	Total	
DSC	BSCCS23501	Big Data Analytics	3	0	0	3	60	40	100	TCS
DSC	BSCCS23502	Big Data Analytics Lab	0	0	2	1	15	10	25	TCS
DSC	BSCCS23503	Artificial Intelligence	3	0	0	3	60	40	100	MD
DSC	BSCCS24504	Research Methodology	3	0	0	3	60	40	100	AK
DSE	BSCCS23505I OT	Internet of Things	3	0	0	3	60	40	100	SP
	BSCCS23505 CG	Computer Graphics								SC
DSE	BSCCS23506I OT	Internet of Things Lab	0	0	2	1	15	10	25	SP
	BSCCS23506 CG	Computer Graphics Lab								SC
VSC	BSCCS23507	Software Project Management	2	0	2	3	60	40	100	SC
MC	MT23CS05A	VU Level - Web Development Using Python	2	0	2	3	60	40	100	PK
TOTAL			16	00	8	20	390	260	650	
Instructions, if any: 1 Theory or 1 Tutorial Hour = 1 Credit, 2 Practical hours = 1 Credit										

BSCCS23503: Artificial Intelligence

Course Type	DSC	Semester	V
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hrs./Week	Lecture	3	CIE Marks	60
Tutorial	0 Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0 Hr./Week	Practical/Studio	-	Total Marks	100
Total	3 Hrs./Week	Total	3		

Course Description

Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems capable of performing tasks that typically require human intelligence. These tasks include learning, reasoning, problem-solving, perception, language understanding, and decision-making. The subject explores various techniques such as machine learning, neural networks, natural language processing, robotics, expert systems, and computer vision. AI aims to build smart agents or programs that can sense their environment, process data, and act intelligently to achieve specific goals.

Course Outcomes

CO No.	Statement
1	Explain basic concepts in AI such as agent and environment
2	Use different search techniques in AI.
3	Represent knowledge using appropriate method.
4	Apply advanced computational techniques in AI.
5	Apply NLP techniques to real world problems.

Mapping of COs to POs and PSOs

OLD

CO Number	POs										PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3	PSO4	
CO1	1	-	-	-	-	-	-	-	-	-	3	1	2	3	Remember
CO2	-	3	-	-	-	-	-	-	-	-	3	1	2	3	Understand
CO3	-	2	-	-	-	-	-	-	-	-	3	1	2	3	Apply
CO4	-	-	3	4	-	-	-	-	-	-	3	1	2	3	Analyze
CO5	-	-	-	-	5	-	-	-	-	-	3	1	2	3	Evaluate

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No – I	Title of Unit - Introduction to Artificial Intelligence	Hours – 6	CO	BTL
	What is Artificial Intelligence , The Foundations of AI, The Historical development of Artificial Intelligence, Applications of AI, AI problems and Techniques, Intelligent Agents- agent, environment, intelligent agents, types of agents, nature and properties of environment		1	2
Unit No – II	Title of Unit – Searching	Hours – 12	2	2,3

State Space, Search and Control Strategies, Problem Characteristics, Issues in Design of Search Programs, Production System. Uninformed Search Techniques: -BFS, DFS, DLS, Iterative Deepening Search, Bidirectional Search, Uniform cost Search. Informed search techniques: - Role of heuristic in problem solving, Hill Climbing, Best First Search, Constraint Satisfaction, Mean-End Analysis, A*, AO*.				
Unit No – III	Title of Unit Knowledge Representation	Hours – 9	3	2,3
Representations and Mappings, Approaches to Knowledge Representation, Knowledge representation methods, Propositional Logic, rules of inference, Predicate logic, Representing Simple facts in Logic, Resolution, Forward and backward chaining, Game Playing- adversarial search, types of games, zero sum game, game tree, Minimax Search algorithm, alpha-beta pruning, Reasoning in AI				
Unit No – IV	Title of Unit – Artificial Neural Networks and Genetic Algorithms	Hours – 9	4	2,3,4
Artificial neural network- components of ANN, ANN architectures, activation functions, Perceptron learning, applications of ANN, advantages and disadvantages of ANN Genetic Algorithm- History and evolution of G.A, Modeling a problem for the application of G.A.— Representation of data in chromosomes, Fitness function, reproduction and convergence, Applications of G.A., advantages of GA, Comparison of ANN and GA				
Unit No – V	Title of Unit - Application Areas of AI	Hours – 9	5	3,4,5
Expert Systems:- Expert problems, characteristics of expert system, components of Expert system, difference between knowledge base and data base, advantages/limitations of expert system, applications of expert system Natural Language Processing:- components of NLP, steps in NLP, applications of NLP-semantic analysis, document classification				

Textbooks

1	<i>Saroj Kausik , Artificial Intelligence, Cengage Learning Publishers</i>
2	<i>Nilsson, Artificial Intelligence: A New Synthesis, Elsevier Publication</i>
3	<i>Ethem Alpaydin, Introduction to Machine Learning, PHI 2nd Edition</i>

Reference Books / Journal Articles / Weblink

1	<i>Elaine Rich, Kevin Knight, & Shivashankar B Nair, Artificial Intelligence, McGraw Hill, 3rd ed.,2009</i>
2	<i>Introduction to Artificial Intelligence & Expert Systems, Dan W Patterson, PHI.,2010</i>
3	http://nptel.ac.in/

BSCCS23504 :: Research Methodology

Course Type	DSC	Semester	V
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hr./Week	Lecture	3	CIE Marks	60
Tutorial	0 Hr./Week	Tutorial	0	ESE Marks	40
Practical/Studio	0 Hr./Week	Practical/Studio	0	Total Marks	100
Total	3 Hr./Week	Total	3		

Course Description

The Research Methodology course in the Bachelor of Computer Science program equips students with essential skills to conduct systematic research in computing and industry-driven projects. It covers fundamental concepts of research, problem identification, and various research designs. Students will explore literature review techniques, data collection methodologies, and statistical tools for data analysis. The course emphasizes technical writing, research documentation, and ethical considerations, preparing students to effectively communicate findings. Through practical applications, students will design and present research projects, aligning with real-world industry challenges. This course fosters critical thinking, analytical skills, and innovation, enabling students to contribute to academic and industrial research domains.

Course Outcomes

CO No.	Statement
1	Recall fundamental concepts of research methodology and problem identification in computing.
2	Explain various research designs, methodologies, and data collection techniques for industry applications.
3	Apply literature review techniques to analyze and formulate research problems effectively.
4	Analyze and interpret research data using statistical and computational tools.
5	Evaluate and document research findings through structured technical writing and proper referencing.
6	Design and present an industry-based research project with real-world applications.

Mapping of COs to POs and PSOs

CO No.	Pos										PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3	PSO4	
CO1	1								3	2				1,2	1
CO2		2	3	1							2	1		2,4	2
CO3			3	2	1						2	1		1,3,4	3
CO4			2	3	1						1	3		5,6	4
CO5		1		3		2	3			2			1	4,5	5
CO6			2	3		3				2			2	6	6

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No. 1	Introduction and Problem Identification	6 Hours	CO	BTL
	Definition and Importance of Research; Types of Research: Fundamental, Applied, and Industry-Based; Research in Computer Science: Trends & Challenges; Identifying a Research Problem: Sources & Techniques; Problem Formulation and Defining Research Objectives; Case Study: Industry Research Problem in Computing		1	1
Unit No. 2	Research Design and Methodology	8 Hours		
	Research Design: Exploratory, Descriptive & Experimental; Qualitative vs Quantitative Research in Computing; Data Collection Methods: Surveys, Interviews, Case Studies; Sampling Methods and Data Validation; Industry-Based Research: Agile, Lean, and Design Thinking Approaches		2	2
Unit No. 3	Literature Review and Hypothesis Formulation	7 Hours		
	Importance of Literature Review in Computing Research; Sources of Literature: Journals, Conference Papers, Patents; Tools for Literature Review: Google Scholar, Scopus, IEEE Xplore; Writing a Review Paper: Summarization & Critical Analysis; Formulating a Hypothesis and Research Questions; Ethical Considerations in Research		3	3
Unit No. 4	Data Analysis & Interpretation	8 Hours		
	Data Preprocessing & Cleaning; Statistical Tools & Techniques for Data Analysis; Introduction to Machine Learning in Research; Software Tools: Python, R, MATLAB, or Excel; Data Visualization Techniques for Industry Applications; Case Study: Data-Driven Decision Making		4	4
Unit No. 5	Technical Writing & Research Documentation	8 Hours		
	Research Paper Structure: IEEE & ACM Formats; Writing Abstract, Introduction, Methodology, Results & Conclusion; Referencing Styles: APA, MLA, IEEE; Plagiarism and How to Avoid It (Turnitin, Grammarly, PlagScan); Writing a Research Proposal & Project Report; Industry Collaboration: Writing White Papers & Technical Reports		5	5
Unit No. 6	Industry Applications & Project-Based Research	8 Hours		
	Research-Based Project Selection & Planning; Industry Use Cases: AI, IoT, Cybersecurity, Blockchain, Cloud Computing; Converting Research into Startups & Patents; Funding & Grant Proposal Writing; Presentation & Communication Skills for Research Findings; Final Project Presentation & Review		6	6

Textbooks

1	<i>Research Methodology: Methods and Techniques</i> , C.R. Kothari and Gaurav Garg , NEW AGE INTERNATIONAL (P) LIMITED, PUBLISHERS, Second, 1990.
2	<i>Fundamental of Research Methodology and Statistics</i> , Yogesh Kumar Singh , NEW AGE INTERNATIONAL (P) LIMITED, PUBLISHERS, 2006
3	<i>Research Methodology: Tools and Techniques</i> , Dr. Prabhat Pandey, Dr. Meenu Mishra Pandey , Bridge Center, 2015.
4	<i>Research Methodology a step-by-step guide for beginners</i> , Ranjit Kumar , Third, 2011.
5	<i>The Craft of Research</i> , Wayne C. Booth , Gregory Colomb , Joseph M. Williams , William Fitzgerald , The University of Chicago Press, 2016

Reference Books / Journal Articles / Weblink

1	arXiv:2405.00040v1 , (accessed on 20/03/2025)
2	https://www.coursera.org/learn/research-methods , (accessed on 20/03/2025)
3	<i>Research Design</i> , John W. Creswell, SAGE Publications India Pvt. Ltd., 2009.
4	https://methods.sagepub.com/ , (accessed on 20/03/2025)
5	https://owl.purdue.edu/owl/ , (accessed on 20/03/2025)

BSCCS23505IOT : Internet of Things

Course Type	Theory	Semester	5
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hrs./Week	Lecture	3	CIE Marks	60
Tutorial	0 Hrs./Week	Tutorial	0	ESE Marks	40
Practical/Studio	0 Hrs./Week	Practical/Studio	0	Total Marks	100
Total	3 Hrs./Week	Total	3		

Course Objectives

This course offers an in-depth study of the fundamentals, technologies and uses of the Internet of Things (IoT). The IoT signifies a transformative approach to how devices, sensors, and actuators interact and exchange data, facilitating the development of intelligent systems that optimize efficiency, streamline operations, and introduce novel services. Learners will acquire practical experience in conceptualizing and deploying IoT solutions across diverse sectors, including smart homes, healthcare, industrial automation, and beyond.

Course Outcomes

CO No.	At the end of this course students will be able to
1	To learn the fundamentals, characteristics, Application areas of Internet of Things.
2	To understand IoT architecture, including its layers, components, communication protocol
3	To Illustrate use of IoT sensors, actuators, micro controllers in IoT system
4	To design and develop an IoT application
5	To analyze real-world IoT case studies, applications across various domains and explore advanced IoT concept

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BT L
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO1 2	PS O1	PS O2	PS O3	PS O4	
CO1	1	1	1		1								1		1	1	1,2
CO2	1	1	1		2								1		1	1	2,3
CO3		2	1		2								1		1	1	2,3
CO4		1	2		2								1	1	2	1	3,4
CO5			2										1				5

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL:Bloom's Taxonomy Level

Course Content

Unit No.01	1 Introduction to Internet of Things	9 hours	CO	BTL
	Definitions Internet of Things (IoT); Application areas of The Internet of Things (IoT); History of IOT; Characteristics of IOT; IOT stack; Requirements and structure of IOT; IoT enabling technologies; IOT challenges		1	1,2
Unit No.02	2 : IoT Architecture	9 Hours		

IoT Architecture;IoT Standards; The major component of IOT Hardware; Role of wired and wireless communication Technologies in IOT ; IoT Communication and Networking protocols and its types ; IoT Connectivity ; Industrial Iot and Consumer IoT		2, 3	3,4
Unit No.03	3 IoT Devices and Hardware	9 Hours	
IOT Sensors; Types of IoT sensors; Sensors Interface; Embedded Systems and Devices in IoT; Microcontrollers in IoT; Arduino, Raspberry Pi, ESP32 Architecture;Actuators in IoT;Working of IoT devices and use of Actuators; Types of Actuators		3, 4	2,3,4
Unit No.04	4 Design and develop an IoT application	9 Hours	
IoT Hardware Selection; Design the architecture and components of the IoT solution; Programming IoT Devices with Arduino/ Raspberry Pi / ESP3; Arduino IDE /Raspberry Pi IDE / ESP32 IDE , Design and develop an IOT application with IoT Device Setup and Programming any one		3, 4	3,4,5
Unit No.05	5 Case Studies IoT Applications and Advanced IoT	9 Hours	
Smart Cities; Smart Buildings; Smart Homes; Smart Car; Smart Healthcare Applications; IoT Data Collection and Storage; Big Data and IoT; Security in cloud for IoT Applications; Data Analytics for IoT Systems		5	4,5,6

Textbooks

1	INTERNET OF THINGS Architecture and Design Principles by Raj Kamal
2	IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things" byDavid Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry
3	Internet of Things: A Hands-On Approach" – Arshdeep Bahga, Vijay Madisetti

Reference Books / Journal Articles / Weblink

Reference Books	
1	Designing the Internet of Things" – Adrian McEwen, Hakim Cassimally
2	Getting Started with the Internet of Things" – Cuno Pfister
3	https://www.techtarget.com
4	https://www.geeksforgeeks.org

BSCCS23506IOT : Internet of Things Lab

Course Type	Theory	Semester	5
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Teaching Scheme		Credits		Examination Scheme	
Lecture	0 Hrs./Week	Lecture	0	CIE Marks	15
Tutorial	0 Hrs./Week	Tutorial	0	ESE Marks	10
Practical/Studio	2 Hrs./Week	Practical/Studio	1	Total Marks	25
Total	2 Hrs./Week	Total	1		

Course Objectives

This course offers an in-depth study of the fundamentals, technologies and uses of the Internet of Things (IoT). The IoT signifies a transformative approach to how devices, sensors, and actuators interact and exchange data, facilitating the development of intelligent systems that optimize efficiency, streamline operations, and introduce novel services. Learners will acquire practical experience in conceptualizing and deploying IoT solutions across diverse sectors, including smart homes, healthcare, industrial automation, and beyond.

Course Outcomes

CO No.	At the end of this course students will be able to
1	To learn the fundamentals, characteristics, Application areas of Internet of Things.
2	To understand IoT architecture, including its layers, components, communication protocol
3	To Illustrate use of IoT sensors, actuators, micro controllers in IoT system
4	To design and develop an IoT application
5	To analyze real-world IoT case studies, applications across various domains and explore advanced IoT concept

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BT L
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PS O1	PS O2	PS O3	PS O4	
CO1	1	1	1		1								1		1	1	1,2
CO2	1	1	1		2								1		1	1	2,3
CO3		2	1		2								1		1	1	2,3
CO4		1	2		2								1	1	2	1	3,4
CO5			2										1				5

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL:Bloom's Taxonomy Level

List of Experiments:

Sr. No.	Lab Assignment	CO	BTL
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1	Infosys Springboard IoT Platforms Overview	1	3
2	Study of Arduino Uno board and Arduino IDE (Integrated Development Environment)	1	4
3	Study of breadboard. Installation of Arduino Software (IDE)	2	4
4	Controlling the LED blink rate with the potentiometer interfacing with Arduino	2	4
5	Interfacing of temperature sensor LM35 with Arduino	3	5
6	Interfacing dht11 humidity sensor with arduino	3	4
7	Study of Raspberry Pi board and IDE	4	4
8	Installing and configuring Raspberry Pi IDE	1	3
9	Controlling IoT LED Using Raspberry Pi and Python	2	4
10	Interfacing of LED, Push button with Raspberry Pi and Python Program	2	3
11	Compare and contrast protocols Wired and Wireless protocols MQTT, CoAP, and HTTP.	3	4
12	Compare and contrast protocols MQTT, CoAP, and HTTP.	4	4
13	Study of Raspberry Pi based Smart Phone Controlled Home Automation	3	4
14	Raspberry Pi based Smart Phone Controlled Home Automation	4	4
15	Interfacing dht11 humidity sensor with Raspberry Pi	5	5

Note: Online platforms can be used for Lab

Textbooks

1	INTERNET OF THINGS Architecture and Design Principles by Raj Kamal
2	IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things" by David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry
3	Internet of Things: A Hands-On Approach" – Arshdeep Bahga, Vijay Madisetti

Reference Books / Journal Articles / Weblink

Reference Books	
1	Designing the Internet of Things" – Adrian McEwen, Hakim Cassimally
2	Getting Started with the Internet of Things" – Cuno Pfister
3	https://www.techtarget.com
4	https://www.geeksforgeeks.org

BSCCS23505CG :: Computer Graphics

Course Type	DSE	Semester	V
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hr./Week	Lecture	3	CIE Marks	60
Tutorial	0 Hr./Week	Tutorial	0	ESE Marks	40
Practical/Studio	0 Hr./Week	Practical/Studio	0	Total Marks	100
Total	3 Hr./Week	Total	3		

Course Description

An introduction to the computer as a graphic design and artist's tool. Using Windows or linux OS, students learn basic use and application of vector illustration, raster image, and scanning software programs to the art and design process. Emphasis is on "hands on" use of the computer, and how the computer can aid the artist's and designer's problem-solving process through interactive visual alternatives. To understand concepts with respect to: Graphics Object, Graphics System. Learn to understand manipulate graphics object. To develop use of Graphics Object.

Course Outcomes

CO No.	Statement
1	To understand how graphics objects are represented in Computer
2	To analyze how graphics system in a computer supports presentation of graphics information
3	To remember how interaction is handled in a graphics system
4	To evaluate graphics object by applying different transformations
5	To apply programmer's perspective of working of computer graphics

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO 9	PO 10	PSO1	PSO2	PSO3	PSO4	
CO1			3		2						2				1
CO2		1		2	1						2		1		2
CO3		3			2						2		1		2
CO4	3		2		1						2		1		2
CO5		3		2							3				6

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Title of Unit: Introduction to computer graphics	Hours 7	CO	BTL
Introduction to computer graphics & graphics systems, Components of Computer Graphics Representation, Presentation, Interaction and Transformations, Applications of Computer Graphic, Pixel/Point, Raster v/s Vector, RGB color model, intensity, Programming essentials			1, 2	1

– event driven programming, OpenGL library.				
Unit No 2	Title of Unit: Input and Output Device	Hours 8	3	2
Logical Interaction – Locator, valuator, pick and choice, Physical devices used for interaction – keyboard, mouse, trackball, spaceball, tablets, light pen, joy stick, touch panel, data glove, Keyboard, Mouse interaction in OpenGL, Presentation Graphics - frame buffer, display file, lookup table, Display devices, Random and Raster scan display devices, Hardcopy devices - Plotters and Printers.				
Unit No 3	Title of Unit: Raster Scan Graphics	Hours 10	2	4
Line drawing algorithms; DDA algorithm, Bresenham's line drawing algorithm, Circle generation Algorithm, Scan conversions- Generation of the Display, Image compression, Displaying Lines and characters, Polygon filling				
Unit No 4	Title of Unit: Transformation	Hours 10	4	4
Basic transformations: translation, rotation, scaling; Matrix representations & homogeneous coordinates, Reflection, shear, Transformation of points, lines, parallel lines, intersecting lines, Viewing pipeline.				
Unit No 5	Title of Unit: Clipping and Hiding Surface Elimination	Hours 10	5	3
clipping operations, point clipping, Polygon clipping, Sutherland Hodgman algorithm, Weiler- Atherton Algorithm, Depth comparison, A-buffer algorithm, Back face detection, Depth –Buffer, Scan-line Method - BSP tree method, the Painter's algorithm, Area-subdivision algorithm.				

Textbooks

1	<i>Introduction to Computer Graphics- David J. Eck, Hobart and William Smith Colleges, David J. Eck, 2016</i>
2	<i>Fundamentals of Computer Graphics by Dr. John Collomosse, CreateSpace Independent Publishing Platform, 2014</i>

Reference Books / Journal Articles / Weblink

1	<i>Computer Graphics (C version 2nd Ed.), Hearn, Pearson education, 2022</i>
2	<i>Computer Graphics using OpenGL, F. S. Hill, Stephen Kelly, PHI Learning, 2015</i>
3	<i>Procedural Elements of Computer Graphics, David F. Rogers, Tata McGRAw Hill, 2017</i>
4	<i>Computer Graphics, Zhigang Xiang, Roy Plastock, McGraw Hill Education, 2006</i>

Course Type	DSE	Semester	V
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Teaching Scheme		Credits		Examination Scheme	
Practical/Studio	2 Hr./Week	Practical/Studio	1	CIE Marks	15
				ESE Marks	10
				Total Marks	25

Course Outcomes

CO No.	Statement
1	To design graphics objects are represented in Computer
2	To understand how interaction is handled in a graphics system
3	To remember how to manipulate graphics object by applying different transformations
4	To evaluate the programmer's perspective of working of computer graphics

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO 9	PO 10	PSO1	PSO2	PSO3	PSO4	
CO1	1		3								3				2
CO2		2			1							2			4
CO3		2										1			3
CO4										1				1	6

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

List of Experiments/ Studio Assignments, etc

Sr. No	Name of Experiment/ Studio Assignment	CO	BTL
1	Design and develop simple graphics programs using basic graphics functions.	1	1
2	Design and develop OpenGL programs to implement basic graphics primitives.	1	1
3	Write a program to simulate any one of or similar object: • Chess / Ludo Board • Mickey Mouse • Analog clock • Tower of Hanoi – A graphical representation	5	6
4	Implement DDA and Bresenham's line drawing algorithm.	2	2
5	Implement Cohen Sutherland line clipping algorithm.	2	2
6	Implement polygon filling algorithms	2	2
7	Write a program to draw 2D object and perform translation, rotation and scaling transformations.	3	2
8	Case study of any graphics tool (Direct3D/Maya/Blender)	2, 5	2. 4
9	Write a program in C to apply 3D transformation on given object	3	6

Textbooks

1	<i>Introduction to Computer Graphics- David J. Eck, Hobart and William Smith Colleges, David J. Eck, 2016</i>
2	<i>Fundamentals of Computer Graphics by Dr. John Collomosse, CreateSpace Independent Publishing Platform, 2014</i>

Reference Books / Journal Articles / Weblink

1	<i>Computer Graphics (C version 2nd Ed.), Hearn, Pearson education, 2022</i>
2	<i>Computer Graphics using OpenGL, F. S. Hill, Stephen Kelly, PHI Learning, 2015</i>
3	<i>Procedural Elements of Computer Graphics, David F. Rogers, Tata McGRAw Hill, 2017</i>
4	<i>Computer Graphics, Zhigang Xiang, Roy Plastock, McGraw Hill Education, 2006</i>

BSCCS23507:: Software Project Management

Course Type	VSC	Semester	V
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Teaching Scheme		Credits		Examination Scheme	
Lecture	2 Hr./Week	Lecture	2	CIE Marks	60
Tutorial	0 Hr./Week	Tutorial	0	ESE Marks	40
Practical/Studio	2 Hr./Week	Practical/Studio	1	Total Marks	100
Total	3 Hr./Week	Total	3		

Course Description

Software Project Management (SPM) is essential for planning, executing, and monitoring software projects effectively. This course introduces students to key concepts like project planning, scheduling, risk and quality management, and team coordination. It covers both traditional and agile methodologies, emphasizing budgeting, resource allocation, and stakeholder communication. By the end of the course, students will be able to develop project plans, track progress using project management tools, and apply risk and quality management strategies. The course includes practical lab assignments and hands-on experience with free tools like Trello, OpenProject, and Gantt charts.

Course Outcomes

CO No.	Statement
1	Understand Software Project Management Concepts
2	Apply Scheduling and Risk Management Techniques
3	Implement Project Monitoring and Quality Assurance
4	Analyze Software Project Metrics and Cost Estimation
5	Utilize Agile and Modern Project Management Tools

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO 9	PO 10	PSO1	PSO2	PSO3	PSO4	
CO1	-	3	-	-	-	2	3	-	-		-	-	-	-	2
CO2	1	-	-	-	-	2	3	-	-	2	-	2	-	-	3
CO3	1	-	-	-	-	2	3	-	1	2	-	-	-	-	4
CO4	1	-	-	-	-	2	3	-	1	3	-	2	-	-	6
CO5	-	-	1	-	-	2	3	-	-	-	-	-	-	-	5

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Title of Unit: Introduction to Software Project Management	Hours 6	CO	BTL
Definition and characteristics of a project Project management processes Role of a project manager Stakeholders Difference between software and traditional project management Introduction to Agile Methodology: Key principles, values, and manifesto			1	2
Unit No 2	Title of Unit: Project Planning and Estimation	Hours 6	2	3
Project scope and requirement analysis Work Breakdown Structure (WBS)				

Empirical estimation models: COCOMO model Agile Estimation Techniques: Story points, planning poker, and relative estimation. Scheduling techniques (Gantt charts, PERT, CPM)				
Unit No 3	Title of Unit: Risk management and Quality Assurance	Hours 6	3	4
Concept of Risk, Type of Risks Risk Mitigation, Monitoring and Management Software Quality Assurance The Software Quality Assurance Plan Software quality models (ISO 9001, CMMI)				
Unit No 4	Title of Unit: Agile Project Management	Hours 6	4	4
Agile vs Traditional methodologies Agile Frameworks Overview: Scrum, Kanban, and Extreme Programming (XP). Agile Project Metrics: Velocity, burnup and burndown charts, and team performance indicators.				
Unit No 5	Title of Unit: Project Execution and Closure	Hours 6	5	3
Team management and communication Monitoring and controlling project progress Financial closure and project reports Case studies of real-world software projects				

Textbooks

1	<i>Software engineering principles and practice, McGraw-Hill, Waman S. Javadekar</i>
2	Agile Project Management for Dummies, 2nd Edition Mark C. Layton, Steven J. Ostermiller
3	Introduction to Software Project Management by Adolfo Villafiorita, CRC Press

Reference Books / Journal Articles / Weblink

1	Bob Hughes, Mike Cotterell, and Rajib Mall, <i>Software Project Management</i> , McGraw Hill.
2	<i>Ian Sommerville, Software Engineering, Pearson.</i>
3	<i>Kenneth S. Rubin, Essential Scrum: A Practical Guide to the Most Popular Agile Process, Addison-Wesley.</i>
4	S.A.Kelkar, "Software Project Management – A concise study", 2nd Ed. PHI learning Pvt. Ltd. 2009 3.
5	Walker Royce, "Software Project Management – A Unified Framework", 2001, Pearson Education.
6	https://learning.tcsionhub.in/
7	www.agilealliance.org
8	http://www.pmi.org
9	https://www.youtube.com/watch?v=SgMOQCe5Ykg

BSCCS23507 :: Software Project Management Lab**Course Content**

List of Experiments/ Studio Assignments, etc

Sr. No	Name of Experiment/ Studio Assignment	Tools to be used	CO	BTL
1	Create a Project Charter for a Software Development Project	Docs	1	2
2	Develop a Work Breakdown Structure (WBS) and Task Network for a project.	GanttProject / OpenProj	2	4
3	Prepare a Gantt Chart and PERT/CPM Chart for project scheduling.	GanttProject / OpenProj	3	3
4	Perform Function Point Analysis for a given software project.	Manual Calculation (Excel)	4	5
5	Identify and analyze risks using a Risk Management Plan.	Excel	5	5
6	Track project progress using Agile methodologies (Scrum or Kanban).	Trello / ClickUp (Free Plan)	3	3
7	Use GitHub for version control and collaboration in a software project.	GitHub	3	3
8	Conduct project closure and create a Project Closure Report.	Docs	1	2

MT23CS05A:: Web Development Using Python

Course Type	Minor Course	Semester	V
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Teaching Scheme		Credits		Examination Scheme	
Lecture	03 Hr./Week	Lecture	2	CIE Marks	60
Tutorial	00 Hr./Week	Tutorial	0	ESE Marks	40
Practical/Studio	02 Hr./Week	Practical/Studio	1	Total Marks	100
Total	03 Hr./Week	Total	3		

Course Description

This course provides a **hands-on introduction** to web development using **Python**, covering fundamental programming concepts, backend development, and web frameworks such as **Flask and Django**. Students will gain practical experience in building **dynamic websites, handling databases, and deploying applications**.

Course Outcomes

CO No.	Statement
1	Understand the fundamentals of Python programming relevant to web development.
2	Develop interactive web applications using Flask and Django frameworks
3	Implement front-end and back-end integration with HTML, CSS, and databases
4	Work with databases and perform CRUD operations using SQL and ORM
5	Deploy Python-based web applications on cloud platforms

Mapping of COs to POs and PSOs

CO Number	Pos										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PS O1	PS O2	PS O3	PS O4	

CO1	3	2	2	1							3	3			Remember
CO2		3	3	3							3	3			Apply
CO3		3	3	2	2						3	3			Apply
CO4		3	3	2	3						3	3			Analyze
CO5			3	3	3							2			Evaluate

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No – I	Title of Unit - Introduction to Python for Web Development	Hours – 06	CO	BTL
Overview of Python programming; Installing Python and setting up a development environment; Basics of Python: Variables, Data Types, Control Statements, Functions; File Handling and Working with Modules; Hands-on: Basic Python programs for web applications			1	2
Unit No – II	Title of Unit – Basics of Web Technologies & Flask	Hours – 06	2	1
Introduction to HTTP, Web Servers, and Web Applications; HTML, CSS, and JavaScript basics for web design; Flask framework: Routing, Request Handling, Templating with Jinja2; Working with Forms and User Input; Hands-on: Building a simple Flask-based website				
Unit No – III	Title of Unit – Database Integration & Backend Development	Hours –06	3	4
Basics of Relational Databases (MySQL, SQLite) ; CRUD Operations: Create, Read, Update, Delete using Python ; Introduction to ORM (SQLAlchemy/Django ORM) ; Handling user authentication and sessions; Hands-on: Developing a web application with database connectivity				
Unit No – IV	Title of Unit – Web Development Using Django	Hours –06	4	3
Introduction to Django framework ; Django MTV (Model-Template-View) architecture ; Creating models, views, and templates; Django Admin Panel & Authentication System; Hands-on: Building a mini-project using Django				
Unit No – V	Title of Unit – Deployment & Cloud Hosting	Hours – 06	5	5

Introduction to **Cloud Deployment Platforms (Heroku, AWS, PythonAnywhere)** ;
 Deploying Python web applications using **Gunicorn & WSGI**; Working with APIs
 and Web Services; Security Best Practices for Web Applications; Hands-on:
Deploying a Flask/Django application to a live server

Textbooks

1	Flask Web Development – Miguel Grinberg
2	Django for Beginners – William S. Vincent

Reference Books / Journal Articles / Weblink

1	Python Crash Course – Eric Matthes
2	Official Flask Documentation: https://flask.palletsprojects.com
3	Official Django Documentation: https://docs.djangoproject.com

Web Development Using Python Lab

Course Content

Sr. No	Name of Experiment/ Studio Assignment	CO	BTL
1	Assignment: Python Basics for Web Development Write a Python program to demonstrate variables, loops, and functions. Implement file handling to read and write data.	1	2
2	Assignment: Building a Simple Flask Web Page Create a Flask web application with multiple routes. Use Jinja2 templating to display dynamic content.	1	1
3	Assignment: Form Handling in Flask Develop a user registration form using Flask. Validate user input and display form responses dynamically.	1	2
4	Assignment: CRUD Operations with Databases Connect a Flask/Django application to a SQLite/MySQL database. Implement Create, Read, Update, and Delete (CRUD) operations.	2	3
5	Assignment: User Authentication System Create a login/logout system using Flask/Django authentication. Store user credentials securely using hashing techniques.	3	5
6	Assignment: Developing a Django Web App	3	5

	Create a Django project with models, views, and templates. Implement a blog or task manager with database storage.		
7	Assignment: REST API Development Build a REST API using Flask/Django REST Framework. Perform GET, POST, PUT, DELETE operations using Postman.	3	5
8	Assignment: Deploying a Web Application Deploy a Flask/Django application to Heroku or PythonAnywhere. Test the deployed application using a live URL	3	5